Shivashant Manohar

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Skills

Programming Languages: Python, TypeScript, JavaScript, Go
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Deep Learning, NLP, Model Evaluation
LLMs and GenAI: Hugging Face Transformers, PEFT, TRL, QLoRA, DPO, RAG, Embeddings, Ollama
Other: Git, GitHub Actions, pytest, Linux, Google Cloud

Projects

NODUS
GitHub | Live

Python, FalkorDB, Cypher, Knowledge Graphs, FastAPI, Next.js

- Built **NODUS**, a company memory that stores post-mortem lessons as role patterns in a **FalkorDB** graph.
- Flags plans shaped like a past failure, even in new words, and names the one step that would repeat it.
- Pre-declared synthetic test: caught **18/18** risky plans, **0/18** false alarms; keyword search flagged all 18.

Cover Song Retrieval
GitHub | Live

Python, PyTorch, NumPy, SciPy, Dynamic Time Warping, pytest, GitHub Actions, Vercel

- Built a music search system that finds cover versions of a song on the **Da-TACOS** benchmark: a **PyTorch** encoder shortlists 30 candidates and a dynamic time warping aligner re-ranks them, cutting query time from **2.63 hours to 38 ms**.
- Improved ranking quality (MAP) **6.8x**, from 0.018 to 0.122, through four controlled experiments, ending within 1.11x of the best system tested.
- Shipped a live demo site backed by **112 tests** and a CI check that keeps the published numbers in sync with the experiment reports.

AgentAlign-Lab
GitHub

Python, PyTorch, Hugging Face TRL, PEFT, QLoRA, DPO, Kaggle

- Built a pipeline that improves a coding agent without human raters: deterministic checkers judge each step it takes in a sandboxed terminal.
- Fine-tuned **Qwen2.5-Coder-1.5B** with **QLoRA** and **DPO** on one free Kaggle T4, raising the pass rate on 42 unseen tasks from **7.1% to 11.3%** and cutting runs with a disallowed command from **41% to 27%**.
- Fixed a first run that showed no gain by matching training prompts to what the agent sees at inference and building pairs from **1,296** of its own rollouts.

Hulu Support Agent
GitHub

Python, Ollama, Qwen2.5, Llama 3.1, RAG, Embeddings, scikit-learn

- Built an AI customer support agent on Hulu's conversations from the Kaggle **Customer Support on Twitter** dataset, running fully on local open-source models.
- Classifies each message into 10 intents, drafts a reply grounded in 3 similar past resolutions, and escalates to a human with a stated reason.
- Scored macro-F1 **0.606** on 200 hand-labelled messages, against 0.369 for keyword rules and 0.355 for TF-IDF.
- Cached every model call to disk so the full evaluation reproduces in **under 20 seconds** with no GPU or API key.

Experience

Agent Orchestrator by Untrivial-ai (12.6k GitHub stars)
September 2026 - Present

Open Source Contributor
Remote

- Opened PR #5638 (+577 code and +694 test lines, under review) that shows each coding agent's **context-window usage** in the session model, CLI and doctor check, so exhausted agents can be respawned.
- Opened PR #5956 (under review) that treats an update-check timeout as a network failure instead of a crash.

Education

VIT-AP University
2023 - 2027

B.Tech in Computer Science (AI and ML), CGPA: 8.05/10
Amaravati, Andhra Pradesh

NBIPU, Bengaluru
Class XII: 94%

Rashtrotthana Vidya Kendra, Dharwad
Class X: CGPA 9.2